

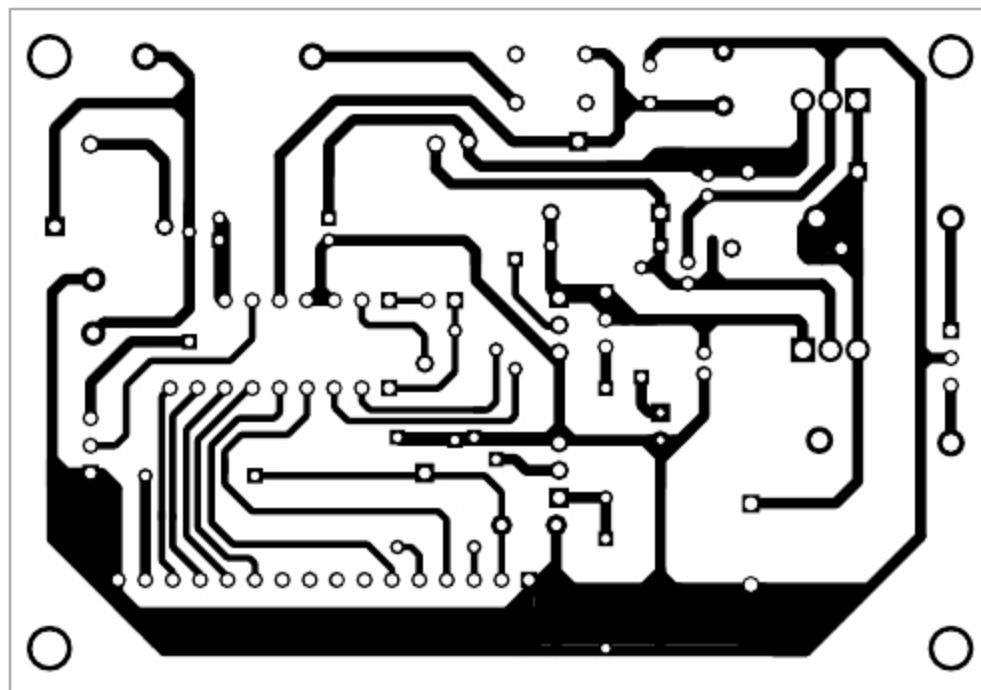


Fig. 2: Actual-size PCB layout of battery charger

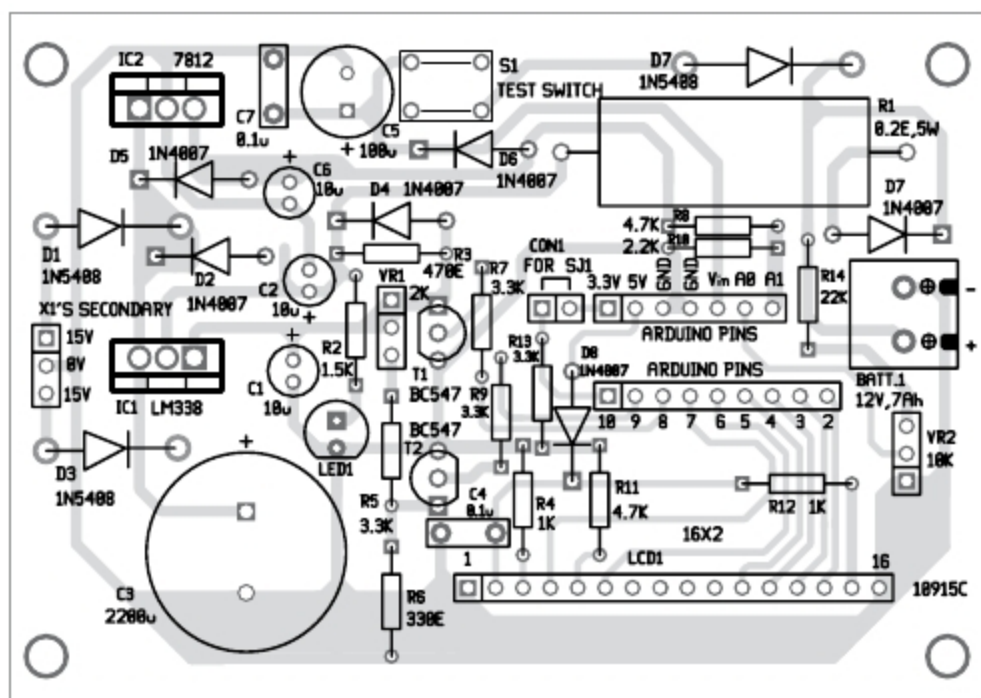


Fig. 3: Components layout for the PCB

to the battery till it is fully charged. Once it is fully charged (above 13V), LM338 supplies about 13.3V to the battery, and maintains the charge at this level. This is called float charging. Charging voltage is controlled by two BC547 transistors (T1 and T2), whose bases are connected to Arduino board pins 8 and 9.

To know the status of the battery when 230V AC mains power is off, press and hold test switch S1 for some time. Arduino gets power from the connected battery. It displays voltage and status of the battery (assuming that battery is not dead).

Software

The software program (batterycharger.ino) is written in Arduino programming language. Arduino IDE is used

to compile and upload the program. ATmega328P on Arduino Uno board comes with a pre-programmed bootloader that allows you to upload a new code to it without using an external hardware programmer.

Connect Arduino to the PC and select the correct COM port in Arduino IDE. Compile the program/sketch. Select the correct board from Tools→Board menu in Arduino IDE. Upload the sketch to the internal memory of the microcontroller.

PARTS LIST

Semiconductors:

- IC1 - LM338 adjustable voltage regulator
- IC2 - 7812, 12V voltage regulator
- LCD1 - 16x2 LCD
- LED1 - 5mm LED
- T1, T2 - BC547 npn transistor
- D1, D3, D7 - 1N5408 rectifier diode
- D2, D4-D6, D8, D9 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

- R1 - 0.2-ohm, 5-watt
- R2 - 1.5-kilo-ohm
- R3 - 470-ohm
- R4, R12 - 1-kilo-ohm
- R5, R7, R9, R13 - 3.3-kilo-ohm
- R6 - 330-ohm
- R8, R11 - 4.7-kilo-ohm
- R10 - 2.2-kilo-ohm
- R14 - 22-kilo-ohm
- VR1 - 2-kilo-ohm potentiometer
- VR2 - 10-kilo-ohm potentiometer

Capacitors:

- C1, C2, C6 - 10 μ F, 25V electrolytic
- C3 - 2200 μ F, 50V electrolytic
- C5 - 100 μ F, 25V electrolytic
- C4, C7 - 0.1 μ F ceramic disk

Miscellaneous:

- Board1 - Arduino Uno board
- X1 - 230V AC primary to 15V-0-15, 1.5A secondary transformer
- BATT.1 - 12V, 7Ah battery
- S1 - Tactile/push-to-on switch
- SJ1 - Shorting jumper
- Jumper wire
- 9-pin Berg strip female connector
- 7-pin Berg strip female connector
- 16-pin Berg strip female connector
- 2-pin terminal connector
- Heat-sinks for IC1 and IC2
- 2-pin male Berg strip connector

Construction and testing

An actual-size PCB layout of the Arduino-controlled 12V battery charger is shown in Fig. 2 and its components layout in Fig. 3.

Assemble the circuit on the designed PCB. Once the circuit is ready and Arduino sketch is written to the board through USB port, connect shorting jumper SJ1 and a multimeter in place of the battery for calibration. Switch on AC mains power and adjust VR1 till the multimeter reads 14.2V DC. Adjust VR2 till LCD1 displays 14.2V. Remove shorting jumper SJ1

and the multimeter. The charger is ready to use. Connect a battery that needs to be charged (12V battery) at BATT.1, as shown in the circuit.

Precaution. Maintain proper polarity of the battery while connecting for charging. Reverse polarity of the battery may damage the circuit.

Charging voltage is set to 14.2V. This value is dependent on the lead-acid battery manufacturer. Hence, refer battery manual before charging.

Use suitable heat-sinks for IC1 and IC2. **EFY**

EFY Note

The source code of this project is available for free download at source.efymag.com



Fayaz Hassan is manager at Visakhapatnam Steel Plant, Andhra Pradesh. He is interested in microcontroller projects, mechatronics and robotics